

2.1 The Cell Membrane

Every living cell is separated from its surroundings by a thin boundary that controls what enters and what leaves. This boundary is not a rigid wall but a flexible, self-sealing film that constantly rearranges itself as the cell grows, divides and responds to signals.

The membrane is built mainly from phospholipids, molecules with a water-loving head and two water-fearing tails. In water they assemble into a double layer with the tails hidden inside and the heads facing outward.

Researchers first described the boundary as a simple film of oil, but careful measurements in the twentieth century showed that it has a defined thickness and a layered structure. Those measurements led to the models that are still taught in introductory courses today.